

NYC Taxi Demand Intelligence Platform

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Cleaned trips
148.4M

Revenue analyzed
\$3.73B

Active days
1,461

Pickup zones
263

Abstract

This report analyzes four years of NYC Yellow Taxi trips and presents a dashboard system built from the processed data. The pipeline cleans 148.4 million valid trips from 2021 through 2024, measures \$3.73 billion in trip revenue, maps demand across taxi zones and borough corridors, and trains a supervised model for zone-hour demand forecasting. Trip volume and revenue grew unevenly between 2021 and 2024: 2024 reached the highest trip volume at 40.4 million trips and \$1.16 billion in revenue, with Manhattan corridors, airport flows, and late-day peaks accounting for a larger share of revenue than at the start of the period. The final product combines a Spark data pipeline, a FastAPI backend, and a Vue dashboard for dispatch planning and route analysis.

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1 Executive Summary

New York City taxi data is large, but the harder business problem is how quickly demand changes by place and time. A fleet operator needs to know where to stage vehicles before demand appears, which corridors produce dependable revenue, and where forecast errors could lead to idle vehicles or long passenger waits. For city planning and investment decisions, the same data can also show where mobility recovery and curb pressure are concentrated.

The analysis supports four main conclusions:

- **Taxi demand recovered unevenly.** Trips increased 28.2% from 2021 to 2022, dropped 3.7% in 2023, and reached a four-year high of 40.4 million trips in 2024. Revenue grew 27.4% in 2023 even though trip volume fell, which is consistent with the December 2022 fare reset being reflected in the 2023 annual data.
- **Average revenue per trip increased.** Revenue per trip rose from \$19.84 in 2021 to \$28.75 in 2024. The increase suggests that fare changes, airport trip mix, and surcharges affected revenue as much as trip volume did.
- **Manhattan accounts for most trips; airport and cross-borough corridors generate higher revenue per trip.** Manhattan-to-Manhattan trips represented 123.3 million trips and \$2.33 billion in revenue. Queens-to-Manhattan routes averaged \$72.68 per trip, compared with \$18.93 for Manhattan-to-Manhattan routes. In 2024, the top 30 pickup zones generated roughly 80% of gross passenger charges.
- **Random Forest provides a useful planning signal.** The best model achieved MAE 5.79, RMSE 14.26, and $R^2 = 0.963$ on the 2024 test period. The dashboard still needs error views because average model scores can hide local misses.

The project combines an offline Spark pipeline, reusable analytical tables, a MongoDB/FastAPI serving layer, and a Vue+D3 dashboard.

2 Business Problem and Decision Questions

Yellow Taxi operations are exposed to a classic supply-demand mismatch. Demand changes by hour, borough, pickup zone, airport activity, day type, and seasonal travel pattern. A driver or dispatcher who reacts only after demand appears loses utilization, increases passenger wait time, and leaves high-value routes under-covered. The same problem matters to city planners because concentrated demand can signal curb pressure, airport access constraints, and neighborhood-level mobility dependence.

This report frames the analysis around five business questions:

1. Where did demand and revenue recover fastest from 2021 through 2024?
2. Which pickup zones and borough corridors generate the most trips and gross revenue?
3. How do hour-of-day and seasonality patterns affect fleet allocation?
4. Can zone-hour demand be forecast accurately enough to support proactive dispatch?
5. How can the analytics be turned into a dashboard that users can inspect repeatedly?

3 Data and Analytical Platform

The data foundation is the NYC Taxi and Limousine Commission Yellow Taxi trip record set from January 2021 through December 2024 [6]. The raw monthly Parquet files are ingested through PySpark,

standardized, cleaned, enriched with taxi zone metadata, and aggregated into analysis tables used by the dashboard and forecasting pipeline. The cleaned dataset contains 148.4 million valid trip records after timestamp, location, fare, distance, duration, and passenger-count validation.

In this report, revenue is measured with TLC total `amount`, which means the gross amount charged to passengers and does not include cash tips. This includes fares, tolls, taxes, fees, and surcharges, so it should be read as a passenger-side gross charge measure rather than fleet or operator net revenue [5].

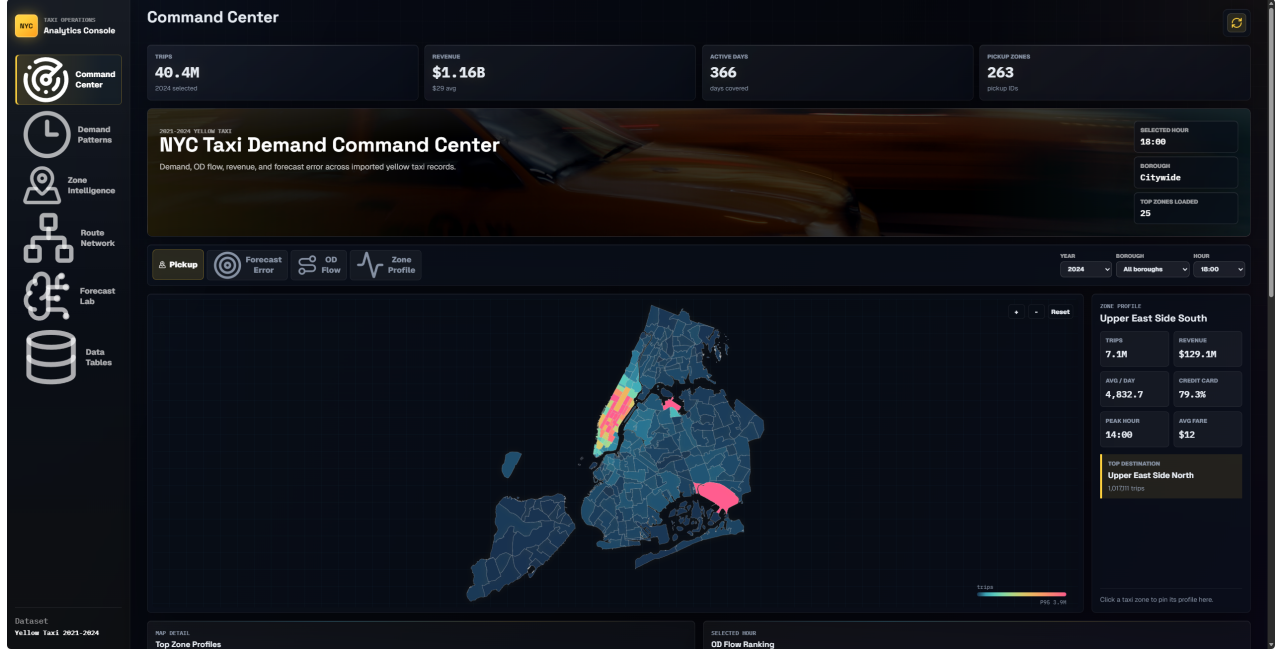


Figure 1: Command Center view with current-year KPIs, map filters, and a selected zone profile.

Platform design

The platform separates heavy compute from dashboard serving:

- **Offline compute:** PySpark jobs process raw Yellow Taxi data, produce partitioned Parquet tables, and export curated CSV outputs.
- **Analytical layer:** temporal, zone, route, airport, profile, and forecasting tables are generated as reusable business metrics.
- **Serving layer:** MongoDB Atlas stores aggregate dashboard collections, not raw trip records.
- **API and dashboard:** FastAPI exposes structured endpoints; Vue and D3 render the dashboard.

4 Demand Recovery and Revenue Trends

Yellow Taxi trips grew from 30.7 million in 2021 to 40.4 million in 2024, with a 3.7% dip in 2023. Revenue grew more steadily, from \$609.1 million to \$1.16 billion over the same period.

Table 1: Annual demand and revenue profile, 2021–2024

Year	Trips (M)	Revenue (\$B)	Avg. revenue/trip	Trip YoY	Revenue YoY
2021	30.7	0.61	\$19.84	–	–
2022	39.4	0.86	\$21.92	+28.2%	+41.7%
2023	37.9	1.10	\$29.01	-3.7%	+27.4%
2024	40.4	1.16	\$28.75	+6.6%	+5.7%

The trend also fits the broader post-COVID recovery in city travel. Taxi trips were still rebuilding in 2021, then rebounded in 2022 as tourism, office trips, and airport travel came back. The 2023 and 2024 numbers reflect changes in trip mix rather than volume alone: 2023 produced fewer trips than 2022 but generated more revenue, and 2024 reached both the highest trip count and the highest annual revenue in the period.

Policy and fare context

A key policy detail is the TLC fare reset that took effect on December 19, 2022 [1]. The rule increased the initial meter charge, per-unit metered fare, rush-hour and nighttime surcharges, improvement surcharge, JFK flat fare, and airport-related surcharges. TLC’s 2022 annual report also framed the increase around pandemic recovery, office return, tourism rebound, air travel recovery, inflation, driver expenses, and a tight labor market [2].

This matters for interpreting 2023. The revenue increase in that year should not be read as demand recovery alone. It also reflects a regulated pricing change that raised passenger charges per trip, especially for metered trips, peak-period trips, airport trips, and Manhattan-related trips.

Two other policy items matter for the revenue fields. The NYS congestion surcharge applies to trips that begin, end, or pass through Manhattan south of 96th Street, with a \$2.50 surcharge for medallion taxicab trips that is passed through to passengers and shown separately [3]. For airport pickups, the Port Authority airport access fee is also added to passenger fares; at LaGuardia, the taxi pickup fee is \$1.75 effective April 5, 2023 [4]. Detailed policy sources are listed in Appendix References [1]–[5].

What drove average revenue per trip from \$19.84 to \$28.75?

Average revenue per trip increased by \$8.91 from 2021 to 2024. The largest visible change is the fare component, which rose by \$6.35 per trip. Tips and other fees also increased, while average trip distance actually fell from 6.92 miles to 4.93 miles. This is an accounting breakdown, not a causal test, but it shows where the dollars moved.

Table 2: Average revenue per trip decomposition, 2021 to 2024

Component	2021	2024	Change
Fare amount	\$13.61	\$19.96	+\$6.35
Tips	\$2.36	\$3.37	+\$1.01
Tolls	\$0.39	\$0.58	+\$0.19
Airport fee	\$0.06	\$0.14	+\$0.08
Other fees and surcharges	\$3.42	\$4.70	+\$1.28
Total revenue per trip	\$19.84	\$28.75	+\$8.91

The decomposition lines up with the policy background. The largest increase came from fare amount, which matches the higher metered fare structure adopted at the end of 2022. The increase in other

fees and surcharges also fits the higher improvement surcharge, peak-period surcharge, airport-related charges, and congestion-related fees on Manhattan trips. Airport recovery still matters as trip mix: airport trips increased from 7.0% of trips in 2021 to 10.1% in 2024, and airport trips averaged \$78.54 in 2024 compared with \$23.16 for non-airport trips. The annual aggregate used here does not split congestion surcharge as its own line, so it is included in other fees and surcharges instead of being estimated separately.

Volume and revenue per trip moved in different directions in 2023, which points to fare changes, fee increases, and airport mix rather than a simple demand rebound.



Figure 2: Demand Patterns view showing monthly demand, hour-of-day demand, seasonality, and weekday-hour structure.

For fleet planning, the hour-of-day curve is especially important. A daily average can hide that the same number of vehicles may be too many in early morning hours and too few during late afternoon and evening peaks. The weekday-hour heatmap gives a more practical scheduling view for staffing, airport queues, and zone staging.

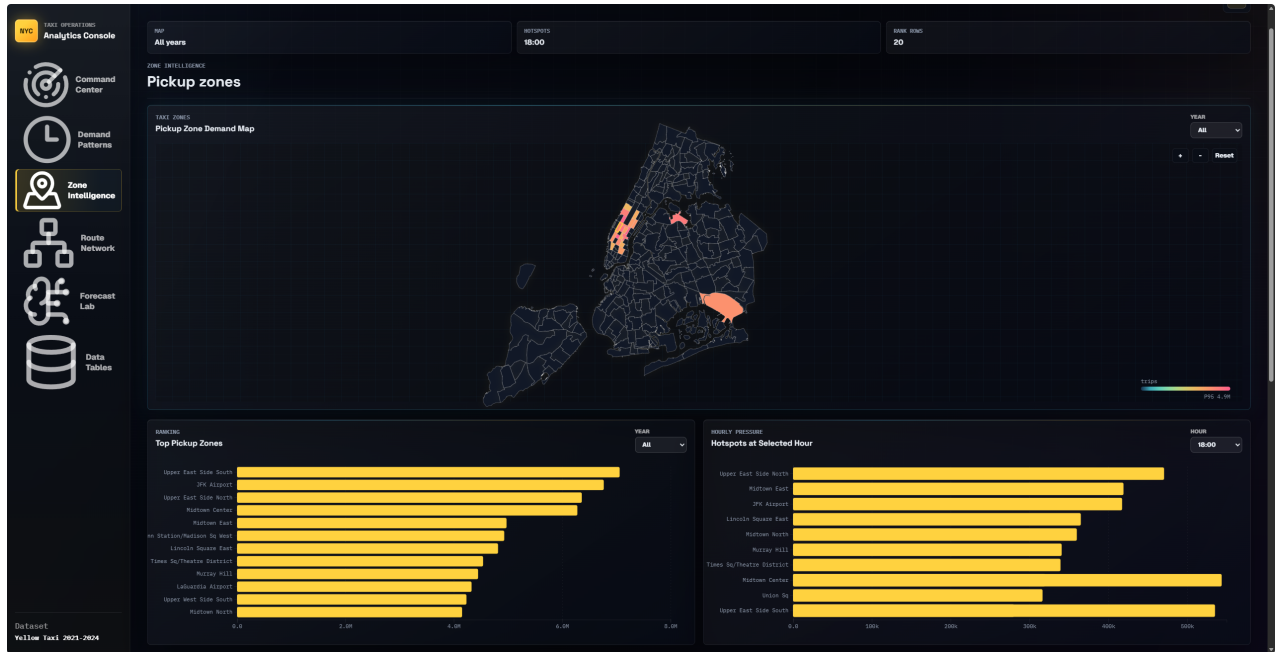
5 Zone Hotspots and Route Network Intelligence

The strongest recurring pickup zones are concentrated in Manhattan, especially Upper East Side, Midtown, Penn Station, Lincoln Square, and Times Square corridors. However, zone movement between 2021 and 2024 shows that demand growth is not static. Times Square/Theatre District improved from rank 19 to rank 6, while LaGuardia Airport improved from rank 22 to rank 9, indicating renewed travel, tourism, and airport access demand.

Table 3: Selected zones with major rank improvement from 2021 to 2024

Pickup zone	2021 rank	2024 rank	Trip change	Trip growth
Times Sq/Theatre District	19	6	642,016	+89.5%
LaGuardia Airport	22	9	594,950	+87.6%
West Village	23	16	299,956	+47.4%
West Chelsea/Hudson Yards	31	24	296,041	+71.4%
Greenwich Village South	36	29	254,185	+76.4%

The dashboard turns these rankings into an interactive view, so users can compare long-term zone importance with hour-specific hotspots. A consistently high-rank zone may need regular staging, while a short-window hotspot may need temporary repositioning or driver incentives.

**Figure 3:** Zone Intelligence view comparing top pickup zones with selected-hour hotspots.

Zone revenue is also highly concentrated. In 2024, the top 10 pickup zones generated 46.0% of gross passenger charges, the top 30 generated 80.0% on a rounded basis, and the top 50 generated 94.3%. More precisely, the 31st ranked zone crossed the 80% threshold out of 263 pickup zones. This gives fleet planning a clearer coverage rule: the highest-value zones deserve planned monitoring and staged coverage, while lower-volume zones can be served more flexibly unless they show temporary spikes.

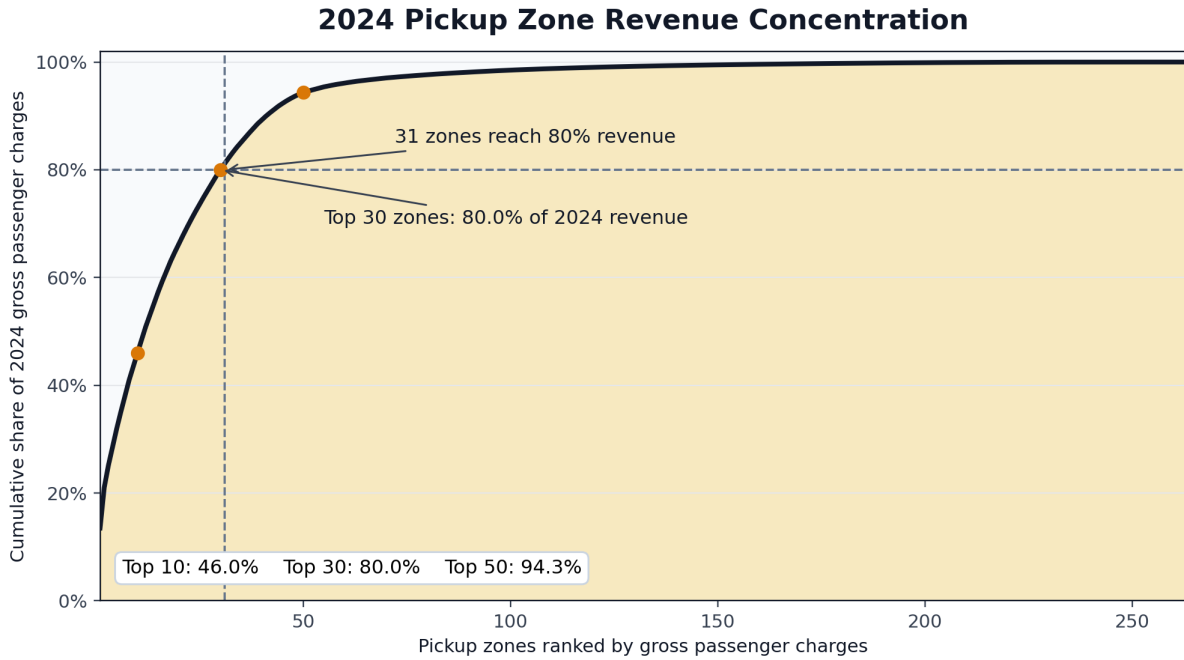


Figure 4: Cumulative 2024 pickup-zone revenue curve. A small set of zones drives most gross passenger charges, which supports a tiered coverage strategy instead of equal monitoring across all zones.

Route-level analysis adds revenue context to the hotspot maps. Dense Manhattan local trips drive most of the volume, while airport and cross-borough routes generate much higher average revenue per trip. Manhattan-to-Manhattan trips account for the largest total revenue pool because of volume. Queens-to-Manhattan and Manhattan-to-Queens corridors are smaller, but they pay much more per trip.

Table 4: High-value borough corridors from pickup-to-dropoff borough aggregates, 2021–2024

Corridor	Trips (M)	Revenue (\$B)	Avg. revenue/trip
Manhattan to Manhattan	123.3	2.33	\$18.93
Queens to Manhattan	7.2	0.53	\$72.68
Manhattan to Queens	4.4	0.26	\$58.09
Manhattan to Brooklyn	3.1	0.12	\$38.85
Queens to Brooklyn	2.0	0.12	\$60.88

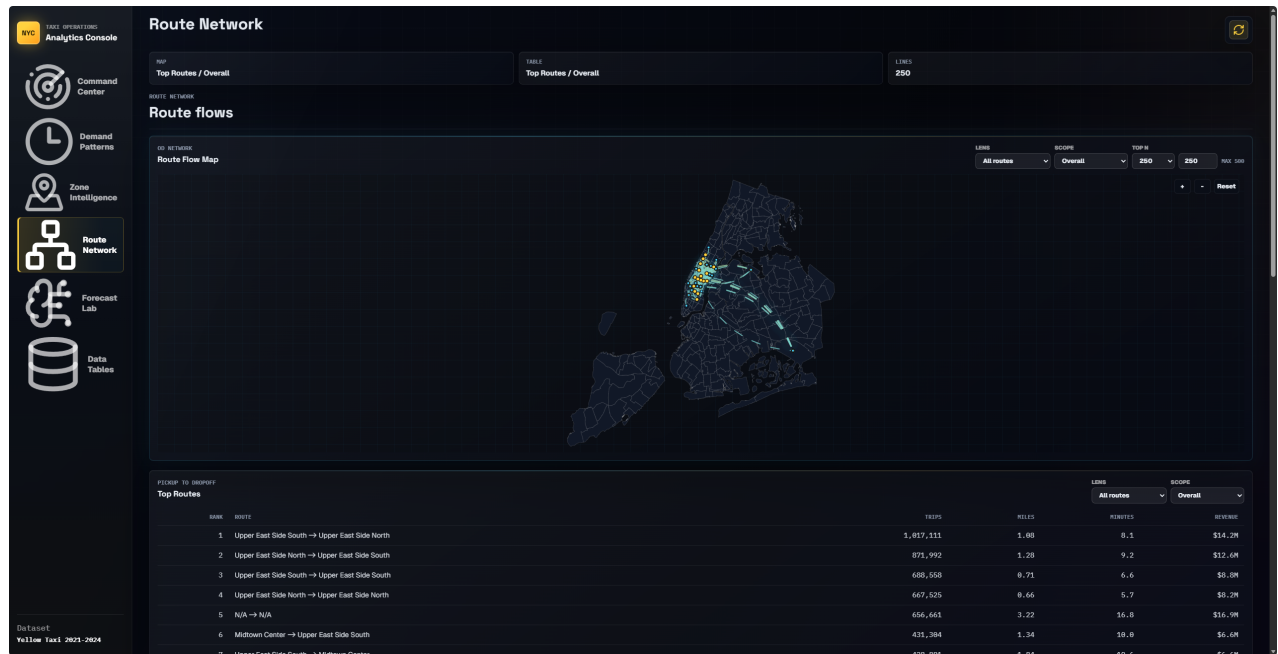


Figure 5: Overall route network view. Short Manhattan routes dominate the map, while longer cross-borough routes appear as separate corridors.

The route screen can switch between overall, inter-borough, and airport views. The overall view is dominated by short Manhattan circulation. The inter-borough view filters out much of that local movement, making Queens, Brooklyn, and airport-adjacent flows easier to read.

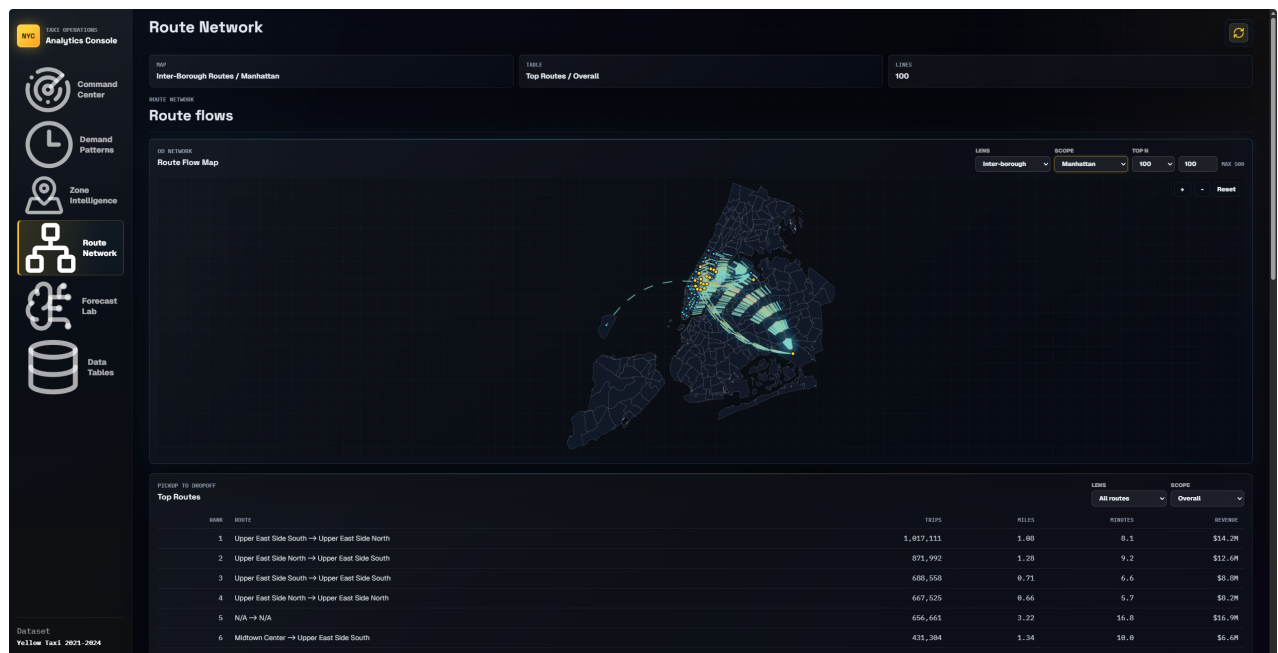


Figure 6: Inter-borough route view focused on Manhattan, with local circulation reduced so cross-borough flows are easier to see.

Airport trips should be analyzed separately because their trip length, revenue profile, queue constraints, and timing patterns differ from regular city-center taxi trips.

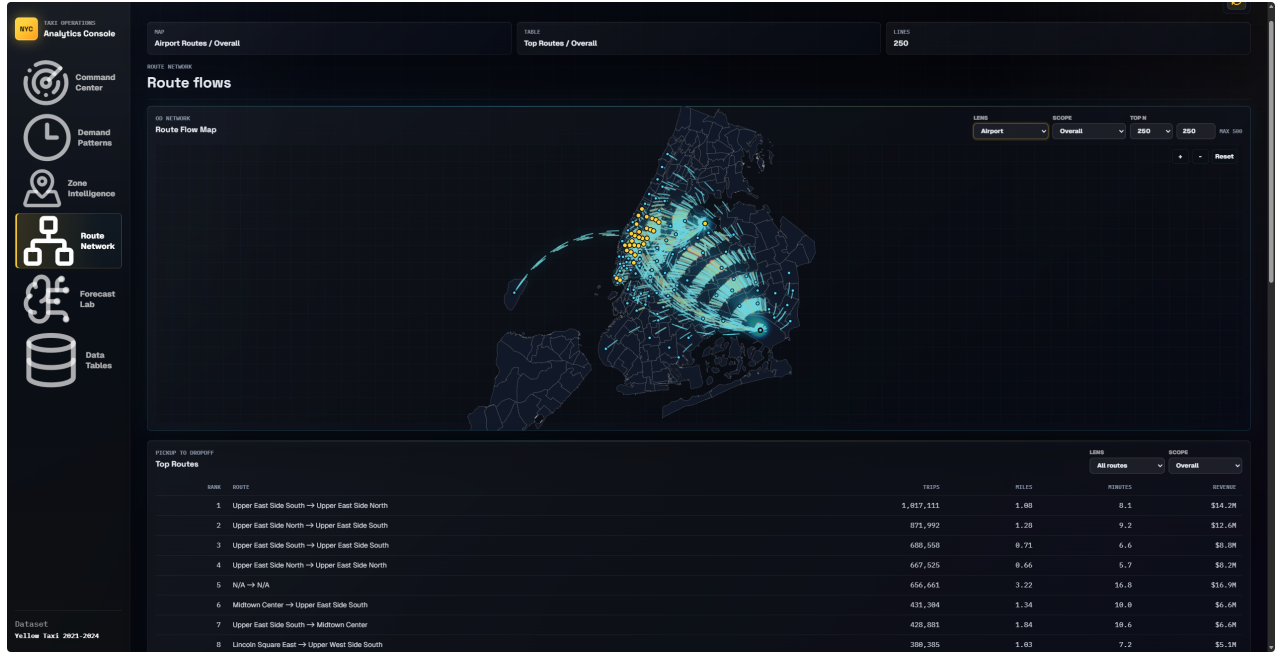


Figure 7: Airport route view highlighting airport-linked corridors with different revenue, queueing, and timing patterns.

Fleet allocation should therefore use route type, not just trip count. Manhattan local service keeps vehicles utilized; inter-borough service covers longer trips; airport service generates the highest revenue per trip. Each segment requires different staging and incentive rules.

6 Forecasting Demand for Operations

The forecasting objective is to predict trip count by pickup zone and hour. The model uses engineered temporal and spatial features, including lagged demand and rolling demand signals. The training design uses 2021–2023 for model learning and 2024 as the evaluation period, which gives a realistic holdout test for forward-looking planning.

Table 5: Model performance on the 2024 evaluation set

Model	MAE	RMSE	R ²
Random Forest	5.79	14.26	0.963
Gradient Boosting	7.17	16.63	0.950
Linear Regression	8.97	19.76	0.929

Random Forest performed best on the 2024 holdout (MAE 5.79, $R^2 = 0.963$). The average miss of about six zone-hour trips is small relative to peak demand in large Manhattan zones but can be material in low-volume zones. For actual use, it should be treated as planning support, not a standalone dispatch engine, and users still need to inspect errors by borough, hour, and zone.

The strong score is reasonable for this task. Taxi demand has stable spatial baselines: Midtown, the Upper East Side, Times Square, and airport zones are repeatedly high-demand areas, while many outer zones stay low-volume. Hour, weekday, and monthly patterns are also regular, and recent demand usually carries into the next few hours. Because the model predicts aggregated zone-hour trips instead of individual passenger behavior, much of the trip-level noise is smoothed before forecasting.



Figure 8: Forecast Lab view with final model scores, actual-versus-predicted demand curves, and error diagnostics for the 2024 test period.

For dispatch use, we also need to inspect where the model makes larger errors. A single RMSE can hide local misses, especially in low-volume zones or during sharp demand peaks. The error map helps users decide where manual review, extra features, or conservative staffing buffers may be needed.

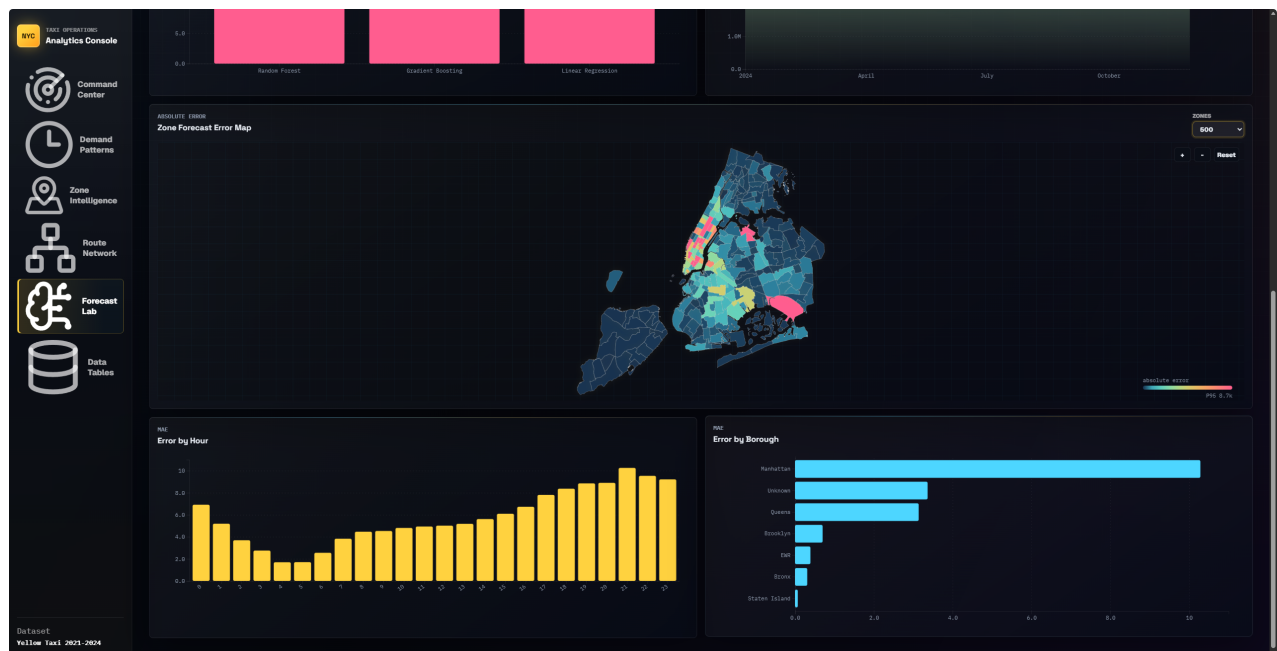
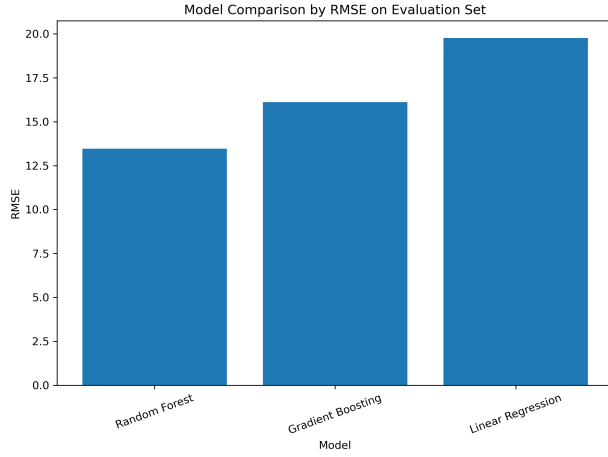
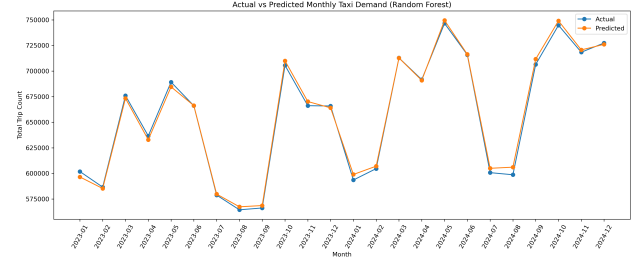


Figure 9: Forecast error map by zone, showing where model misses are larger than the overall score suggests.

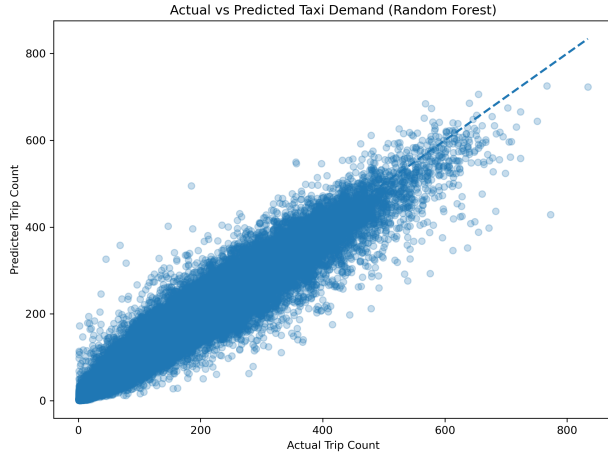


(a) Model comparison by RMSE.

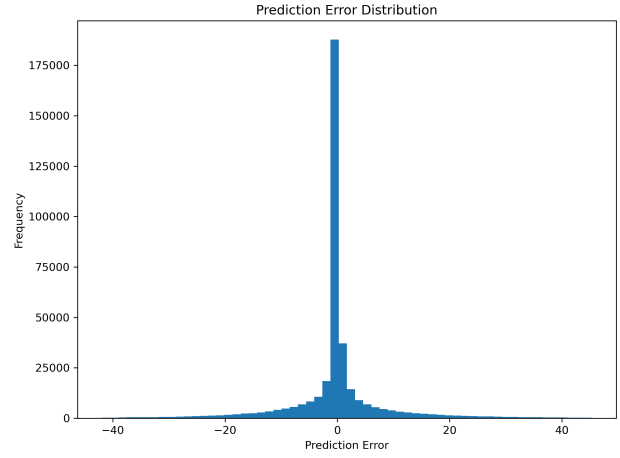


(b) Monthly actual versus predicted demand.

Figure 10: Model comparison and monthly prediction curve used to choose the final forecasting model.



(a) Actual versus predicted scatter.



(b) Prediction error distribution.

Figure 11: Prediction scatter and error distribution for checking where the forecast tracks actual demand and where it misses.

7 Dashboard Productization

The dashboard is the main interface through which users inspect processed metrics, route patterns, and forecasting results:

- **Command Center:** executive KPI strip, map explorer, zone profiles, selected-year and selected-hour context.
- **Demand Patterns:** monthly, hourly, seasonal, and weekday-hour views for staffing and supply planning.
- **Zone Intelligence:** pickup zone rankings and hotspot comparisons for dispatch decisions.
- **Route Network:** origin-destination flows, borough movement, airport routes, and corridor concentration.

- **Forecast Lab:** model performance, actual-versus-predicted curves, and error diagnostics.

The borough filter changes the scale of the analysis from citywide patterns to local movement. In the Brooklyn drill-down, the OD flow map focuses on Brooklyn-relevant trips at 18:00 and shows an all-period zone profile for Downtown Brooklyn/MetroTech: 110.9K trips, \$3.2 million in revenue, a 17:00 peak hour, and a \$23 average fare. This view is useful because Brooklyn demand does not follow the same pattern as the Manhattan core.

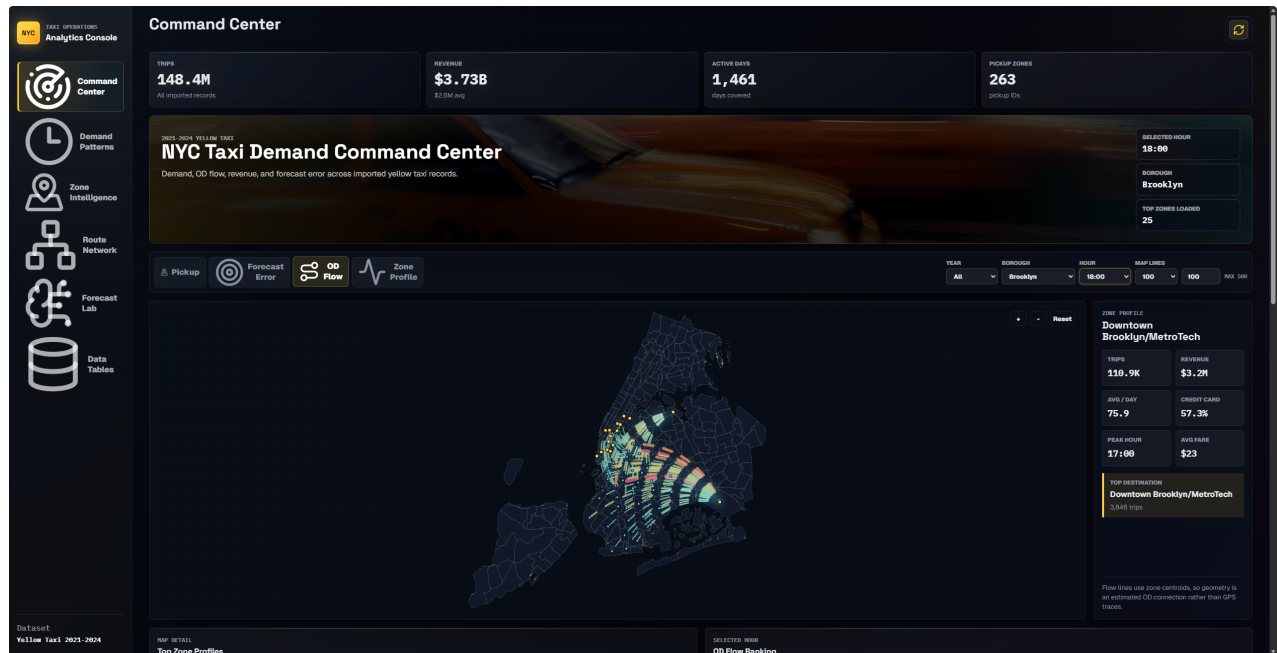


Figure 12: Brooklyn drill-down with OD flow filtering and a selected profile for Downtown Brooklyn/MetroTech.

The dashboard also includes table views for users who want the numbers behind the charts. The payment-type table is a simple example: it lets users inspect payment behavior and revenue mix directly instead of relying only on visual summaries.

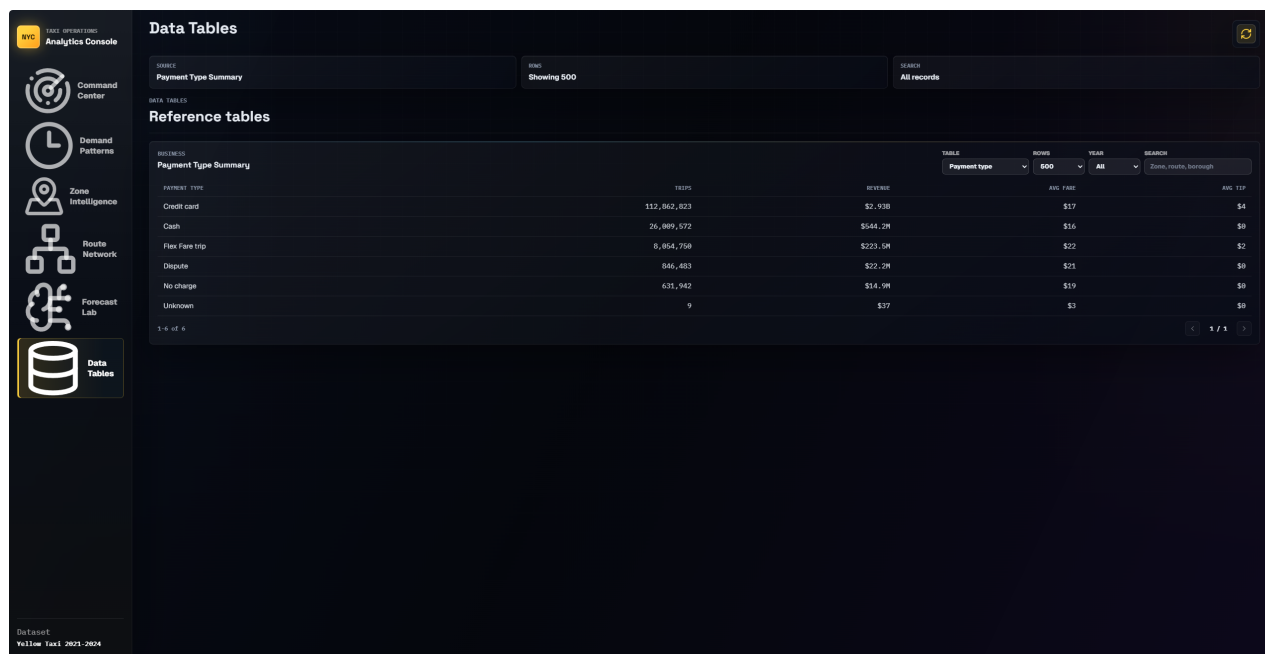


Figure 13: Payment-type table with the numeric breakdown behind payment behavior and revenue mix.

8 Business Recommendations

1. Stage vehicles before late-day demand peaks

In 2024 weekdays, the top five Manhattan pickup zones had 79.8% more combined trips per zone-hour during 16:00–19:00 than at 08:00. The largest zone-level increases were Midtown Center at 237.9%, Times Sq/Theatre District at 126.8%, Midtown East at 84.9%, and Upper East Side South at 68.1%. The recommendation is to move vehicles into these zones before the late-day peak, not after the map has already turned hot. This should start with the highest-value zone set: as shown in Figure 4, the top 30 pickup zones generated roughly 80% of 2024 gross passenger charges. As a scale check, if 100 vehicles each captured one additional late-day trip on 260 weekdays, the gross annual revenue opportunity would be about \$747K at the 2024 average of \$28.75 per trip, before driver payout and repositioning cost.

2. Treat Manhattan circulation and airport flows as different route types

Manhattan-to-Manhattan trips dominate volume with 123.3M trips and \$2.33B in revenue, but they average only \$18.93 per trip. Queens-to-Manhattan routes average \$72.68 per trip, and Manhattan-to-Queens routes average \$58.09. Airport trips also averaged \$78.54 in 2024, more than three times the non-airport average of \$23.16. A simple scale check shows why this is not only a revenue-per-trip question: four Manhattan-local trips at \$18.93 each gross about \$76, roughly the same as one 2024 airport trip at \$78.54. The two segments look similar before costs at that simple hourly scale, but they carry different operating risk. Manhattan service depends on keeping short trips turning over; airport service depends more on queue time, empty return time, and trip-length variance.

3. Monitor fast-rising zones as early demand signals

Times Square/Theatre District gained 642K trips from 2021 to 2024, an 89.5% increase, and moved from rank 19 to rank 6. LaGuardia Airport gained 595K trips, an 87.6% increase, and moved from rank 22 to rank 9. Times Square was already rank 10 by trip count in the first half of 2022 and finished 2022

at rank 8, so the shift appeared well before the 2024 year-end ranking. As a sensitivity check, if weaker staging captured just 1% fewer Times Square trips during 2022–2023, that would equal about 24.6K trips. Using Times Square’s actual 2022–2023 average of \$25.31 per trip, the scale is about \$0.62M in gross passenger charges; using the 2024 citywide average of \$28.75, it would be about \$0.71M. This is not a measured loss; it is a scale estimate showing why rank movement should be tracked alongside static top-zone lists.

4. Use forecast error as a management metric

The Random Forest model has MAE 5.79 trips per zone-hour, which is useful for planning but still large for low-volume zones. For example, a six-trip miss is small if a zone-hour expects 80 trips, but it is large if the forecast is only 12 trips. Forecast accuracy should therefore be reviewed by borough, hour, and zone, not only by the overall RMSE. Larger errors can point to cases where the model needs features such as weather, events, transit disruption, or flight schedules.

9 Limitations and Future Work

Several gaps remain for future work:

- **External context is not yet modeled.** Weather, event schedules, airport flight volume, and traffic incidents are not included.
- **Competitive landscape is not modeled.** Yellow Taxi operates within a broader for-hire vehicle market that includes rideshare platforms such as Uber, Lyft, and Via. This report focuses on optimizing Yellow Taxi operations within observed Yellow Taxi demand, but it does not analyze market share or cross-platform substitution. Incorporating TLC High Volume For-Hire Vehicle trip records would allow the dashboard to compare where Yellow Taxi retains an advantage versus rideshare.
- **Move to a real production environment.** The current pipeline can be moved from local execution to YARN/HDFS or cloud storage so the same Spark jobs can run as scheduled batch workloads.
- **Add real-time data for streaming analysis.** Kafka and Spark Streaming could ingest real-time taxi, traffic, or event data and update demand patterns continuously instead of relying only on offline monthly or daily batch outputs.
- **Scale model training with MLlib.** Distributed training would make it easier to test larger feature sets and retrain models as more months or external signals are added.

10 Appendix A: Technical Documentation and GitHub Reference

The full technical implementation is documented in the project repository:

<https://github.com/bigdata-26spring/NYC-Taxi-Trip-Analytics>

App link: <https://nyc-taxi-trip-analytics.vercel.app/>

The README serves as the technical reference for:

- raw data expectations and TLC Yellow Taxi source files;
- Spark ingestion, cleaning, feature engineering, and analytics scripts;
- forecasting data preparation, model training, and evaluation outputs;
- MongoDB export workflow for dashboard-ready aggregate collections;

- FastAPI backend routes and dashboard sections;
- local execution, Render backend deployment, Vercel frontend deployment, and cluster-mode scaling notes.

11 Appendix B: References

- [1] NYC Taxi and Limousine Commission / NYC Rules. [Taximeter Rate of Fare and Various Surcharges](#). Accessed May 11, 2026.
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